Dysthe Solitary Waves and the Published Tests

Literature review and computational audit

27 October 2026; revised after the cluster-results push

Abstract

The defensible research target is to determine which resolved water-wave Dysthe solitary profiles remain close to their symmetry orbits over a measured propagation interval, and whether neural approximations reproduce that behavior. Two September 2026 preprints refine the theoretical context, but neither supplies an orbital-stability theorem for the implemented periodic free-surface model. The newly pushed cluster snapshot reports qualified periodic references through $\xi = 64$, but none of 24 evaluated neural fits passes the combined $\xi = 4$ gates; 12 random-field fits have execution errors. Detailed reference and failure records remain unpublished. A fresh projection audit shows that copying the generic-packet neural cutoff to the solitary-wave domain would impose a 66.1% relative H^2 error before optimization. This report distinguishes literature results, published measurements, local historical evidence and proposed tests.

1 The question and the exact equation

The project's original question is whether solutions remain smooth over increasing propagation distance. Solitary waves provide a controlled family with which to investigate this question, but existence of one traveling profile does not establish global existence for arbitrary initial data.

The active model evolves a scaled complex *free-surface envelope* $u(\xi, \tau)$ in downstream distance ξ, with retarded-time coordinate τ:

$$\begin{aligned} u_\xi = F(u) = & -iu_{\tau\tau} - i|u|^2 u \\ & - \epsilon(8|u|^2 u_\tau + 2u^2\overline{u_\tau} + 2iu|D_\tau|(|u|^2)). \end{aligned} \tag{1.1}$$

This is the spatial Fedele–Dutykh free-surface convention [1]. Its model identifier is `water-wave-dysthe-spatial-v1`. The Fourier multiplier of $|D_\tau|$ is $|k|$ and that of ∂_τ is ik; thus $\mathcal{H}\partial_\tau = |D_\tau|$. The potential-envelope equation, a temporal equation, and a two-dimensional model are different contracts. Their results cannot be imported by name alone.

Definition 1.1. A traveling solitary envelope on the line has the form

$$u(\xi, \tau) = e^{-i\mu\xi} Q(\tau - s\xi), \qquad Q(z) \longrightarrow 0 \quad \text{as } |z| \longrightarrow \infty. \tag{1.2}$$

The profile satisfies $F(Q) + i\mu Q + sQ' = 0$. A large periodic box approximates this setting only after a separate domain-sensitivity check.

At $\epsilon = 0$ and $s = 0$, the focusing NLS seed is $Q_0(z) = \sqrt{2\mu}\,\mathrm{sech}(\sqrt{\mu}z)$, $\mu > 0$. At nonzero ϵ it is generally only an initial guess. Both the complex phase and shape must satisfy the full Dysthe profile equation. The solitary object is an envelope of many carrier crests, rather than one permanently isolated surface crest.

2 What the current literature establishes

Direct numerical precedent. Fedele and Dutykh constructed spatial Dysthe traveling waves with a Petviashvili iteration and studied their collisions numerically [1]. Their observed inelastic interactions are an established benchmark, not a new discovery to claim from this project. Convergence of that profile iteration is not dynamical stability. Earlier perturbative work by Akylas described how higher-order modulation changes an initially NLS-like solitary envelope, including amplitude and speed adjustments in the studied regime [2]. This motivates comparing an exact NLS inlet with a corrected Dysthe profile.

A 2026 existence result, with a coefficient restriction. Esfahani, Kairzhan and Karazym's preprint, revised 21 September 2026, proves parameter-dependent traveling-wave existence and nonexistence for a general nonlocal derivative NLS family [3]. In their convention,

$$iu_t - u_{xx} - b|u|^2u + i\alpha|u|^2u_x + i\beta u^2\bar{u}_x + \gamma u\partial_x\mathcal{H}(|u|^2) = 0.$$

Matching (1.1) gives

$$(b, \alpha, \beta, \gamma) = (1, 8\epsilon, 2\epsilon, -2\epsilon).$$

Their main existence theorems assume $\gamma \geq 0$; their other negative-γ existence regime requires a vanishing reduced quintic coefficient, whereas ours is $B = -35\epsilon^2/4$. Hence those results do not directly prove existence for this branch. They also explicitly leave stability and instability unaddressed. This is a useful analytical framework, not a certificate for the current solver.

A sharper 2026 obstruction to NLS-type integrability. Sutherland and Yim's 19 September 2026 preprint computes resonant six-wave scattering in Hamiltonian Dysthe models [4]. Its spatial free-surface case is connected to our coefficients by the Fedele–Dutykh gauge transformation, including the induced quintic term. Under its stated formal-functional and regularity assumptions, the result excludes an infinite conserved hierarchy with independent regular C^3 quadratic leading symbols beyond $\mathrm{span}\{1, k, k^2\}$. It does not exclude every conceivable notion of integrability, solitary profiles, or stable propagation. The analysis uses continuous Fourier variables on the line; it is not a periodic-box stability theorem. Both 2026 papers are preprints, not verified journal publications in the records inspected.

Coherent events are broader than exact solitary waves. Schober and Islas's 2025 study uses Floquet spectra to examine soliton-like rogue events in conservative and dissipative higher-order NLS settings [5]. Its publisher abstract was accessible; its full text was not inspected here. Tikan et al. connect random narrowband water-wave fields, coherent-pulse signatures and nonlinear dispersion measurements with NLS/Dysthe/Euler comparisons [6]. These support later random-input diagnostics, not a claim that every large peak is a soliton. The supplied 2026 Schober–Islas preprint concerns weak viscosity and mean-flow damping [7]; such terms require a separately versioned extension, and are not part of (1.1).

3 Four claims that must remain separate

1. **Profile existence:** a nontrivial Q solves the traveling-wave equation. A small computed residual is numerical evidence; an existence theorem needs an analytical or validated numerical argument.

2. **Spectral stability:** the correct linearized evolution has no growing modes in a specified function space. A finite matrix calculation must be checked against resolution and domain changes.

3. **Finite-distance robustness:** selected small perturbations stay close to the traveling-wave orbit over the tested interval. This can be measured with resolved simulations.

4. **Global orbital stability or global regularity:** a uniform statement for all sufficiently small perturbations, or continuation for general data at arbitrary distance. The present tests do not establish either statement.

The phrase *soliton* is used variably in this literature. Here, "solitary wave" describes a persistent localized traveling profile; "integrable soliton" additionally invokes the special scattering structure of an integrable model. Nonintegrability does not imply that every solitary wave is unstable. Likewise, a transient focusing event or a breather need not be a traveling profile.

The small-data global well-posedness and scattering theorem of Mosincat, Pilod and Saut concerns a Dysthe equation on $\mathbb{R}^2$ [8]. Its dimensional and evolution setting differs from this one-coordinate periodic spatial solver. It cannot be presented as a theorem covering our solitary profiles or arbitrary large-amplitude data. No directly applicable orbital-stability theorem for the exact implemented model was found in the sources reviewed; this is a literature search conclusion, not a proof that no such result exists.

3.1 What "long term" means operationally

There is no universal terminal value of ξ, amplitude or H^2 norm that establishes long-term stability. Record the actual interval together with natural propagation scales. With $A_0 = \|u(0)\|_\infty$, the current campaign uses

$$N_{\mathrm{nl}}(\Xi) = A_0^2 \Xi$$

as an initial-peak nonlinear-distance convention. For a traveling wave also record $|\mu|\Xi/(2\pi)$ phase cycles and the shift $s\Xi$ relative to its width. For the stored solitary profile, $A_0 = 1.958116$: the completed $\Xi = 0.1$ check covers 0.3834 nonlinear lengths, whereas the proposed $\Xi = 20$ study would cover 76.68. This is a meaningful increase in scope, not an existing result.

In the model normalization, $\xi = \epsilon^2 k_0 X$ and $\tau = \epsilon(2k_0 X - \omega_0 T)$. Thus physical distance is $X = \xi/(\epsilon^2 k_0)$ once the carrier is specified. The approximation need not remain physically accurate at arbitrarily large X, even if its numerical solution stays smooth. Track effective steepness and spectral bandwidth; use water-wave experiments or higher-fidelity water-wave models for physical validation.

4 Audit of the pushed tests and recorded results

The inspected remote workstream is `codex/long-distance-a100`, at commit

`723883f27a6b3489bfc36f55d9d15b52e8635ad0`.

This new push adds 29 result files to the previously reviewed code commit `c04b44ba`. It includes a `gpu2` cluster snapshot as well as the earlier CPU results. The 39-test software check remains applicable to the unchanged source; it is distinct from the scientific accuracy gates. The new snapshot and its limits are examined below. The earlier report's statement that no cluster results were published is superseded by this revision.

4.1 The CPU pilot supports slabs, with limited scope

The completed pilot used 18 initial fields: six each of packets, modulated backgrounds and smooth random Fourier fields. Twelve trained the operators; three validated and three tested them. PINNs were fitted separately per inlet. Two optimization seeds give six test case–seed pairs, but only three independent test fields. All methods used a small budget; FNO rollout and a per-inlet PINN solve have different inference tasks and must not be treated as identical services.

Method	L^2 (%)	H^1 (%)	Peak (%)	Pairs passing
Global PINN	1.995	3.041	0.936	2/6
NLS-warm PINN	2.074	3.560	1.285	3/6
Four-slab PINN	0.963	1.881	1.089	6/6
FNO	58.739	58.843	32.698	0/6
PINO	56.617	56.833	34.592	0/6

Table 1: Published CPU measurements: worst saved-station relative errors through $\xi = 2$. Gates were 1% complex L^2, 5% H^1 and 5% peak error. No phase or translation alignment was applied. Source: archived `data/cpu-errors.csv`.

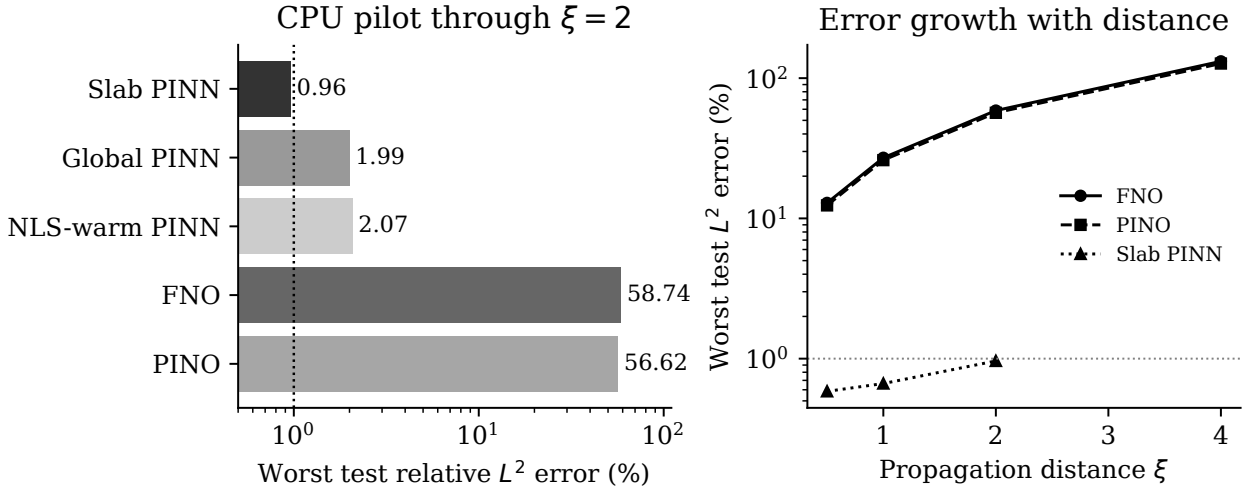


Figure 1: Replot of recorded CPU results. The slab PINN was fitted over each interval; the operators were rolled out beyond their short training windows. These are neither solitary-wave tests nor A100 measurements.

4.2 What the existing results imply for method selection

The slab PINN is the most promising *tested neural method* for the current per-inlet task. It is not yet the fastest solver. Recorded median fitting time was about 1.26 seconds per inlet, versus about 0.0294 seconds for a coarse numerical solve through $\xi = 2$. The latter had worst test H^1 error about 4.62×10^{-7}. These preliminary CPU timings were not an isolated, matched-accuracy benchmark; some training overlapped and setup/audit costs were excluded. They do not establish neural speedup.

The trained FNO and PINO are not reliable long-distance predictors: neither passed all test gates even through $\xi = 0.5$. Their $\xi = 4$ errors exceeded 100%. Twelve operator-training inlets and a small budget are insufficient to draw a general conclusion about FNO, but these particular checkpoints should not be promoted as a solver. NLS warm-starting also failed to improve the aggregate PINN result under this budget.

Three numerical validation fields were independently refined through $\xi = 16$; recorded sampled H^2 norms remained finite, with maximum approximately 4.224. These are finite-distance periodic observations. The new cluster snapshot extends the recorded numerical coverage below; neither set establishes solitary-wave stability or continuum existence.

4.3 The pushed long-distance campaign is a sound staged design

Stage	Distances ξ	Purpose and required observation
Reference	4, 8, 16, 32, 64	Independently refine grid and step; track H^2, spectral tails, action and the NLS control. Stop an unresolved inlet.
Slab PINN	4, 8, 16	Require saved-station complex $L^2 \le 1\%$, $H^1, H^2 \le 5\%$ and peak error $\le 5\%$. Stop later fits after failure.
Sensitivity	4, 8, 16	Check signed response with perturbation halving and independent reference refinements; 1% response agreement.
Packet domain	4, 8, 16, 32, 64	Compare the same analytic localized inlet on two periods; require initial agreement and 1% H^1 domain agreement.

The production design has six validation and six test inlets across three families, with neural seeds 11, 29 and 47. Each PINN is fitted anew; the 4/8/16 horizon fits retrain early slabs independently. Training-reserved inputs are not used to train an FNO in this campaign. A100 access therefore scales the selected per-inlet experiment; it does not already implement operator generalization.

The published tiny workflow completed 24 case/stage tasks. One PINN failed its accuracy gate at $\xi = 0.2$, both packet-domain tests failed their coarse-domain criteria, and four domain tests were correctly marked inapplicable. All six reference and sensitivity checks completed. These retained failures are useful: workflow completion is not "all scientific gates passed." Replaying completed fits preserved 180 artifact hashes; this is CPU reuse evidence, not CUDA restart equivalence. Allocation-specific CUDA checks remain essential.

4.4 Newly published cluster results: the actual outcomes

The new `results/long-distance-gpu2-2026-10-26` snapshot changes the evidence status. Its configuration matches the production design. Its manifest identifies source commit `c04b44ba`; all eight recorded source hashes match the previously tested pinned sources. All 28 published metadata files match their manifest hashes and byte lengths; the manifest is the 29th file.

Stage	Recorded result in the new summary
Periodic reference	12/12 qualified through $\xi = 64$.
Numerical sensitivity	12/12 qualified through $\xi = 16$.
Packet domain	Four pass $\xi = 4$, fail $\xi = 8$; eight nonpackets inapplicable.
PINN accuracy	0/24 evaluated packet/sideband fits qualify through $\xi = 4$.
PINN execution	12 random-field fits: execution errors, no accuracy results.
Later PINN horizons	No evaluated $\xi = 8$ or 16 results; campaign incomplete.

Each validation/test split has 12 of 18 requested case–seed pairs evaluated, representing four of six independent inlets. Both evaluated families are represented, but the random family is absent. The following are therefore *incomplete-subset* statistics; failed executions cannot be counted as accurate predictions or omitted from the coverage statement.

Split/statistic	L^2 (%)	H^1 (%)	H^2 (%)	Peak (%)
Validation worst	1.968	5.826	30.741	1.383
Validation median	1.347	3.419	14.681	0.820
Test worst	1.577	4.130	22.115	1.498
Test median	0.934	2.968	14.900	0.669
Acceptance gate	1.000	5.000	5.000	5.000

Table 2: Recorded interval errors through $\xi = 4$. Medians are across the 12 evaluated case–seed maxima in each split, not across all requested inputs. Detailed H^2 traces were not included in the pushed snapshot.

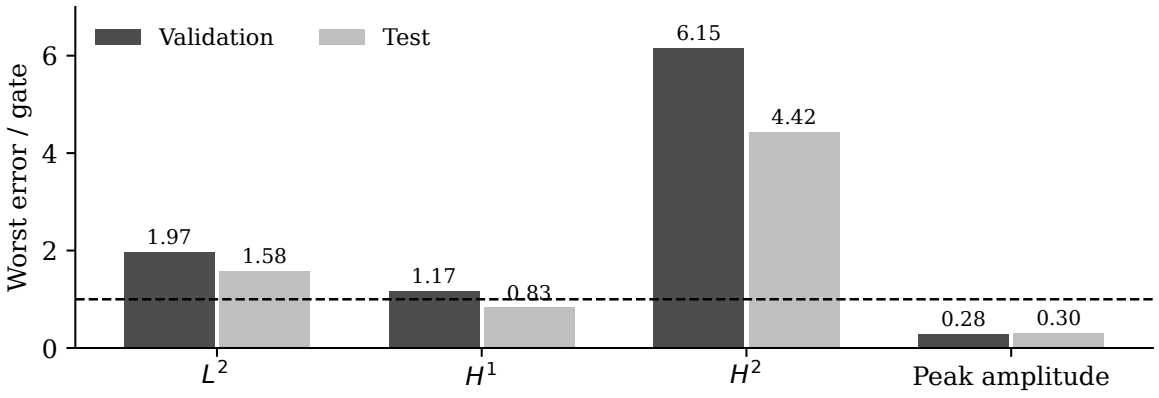


Figure 2: Worst reported errors divided by their acceptance thresholds. A value above one fails that metric. The derivative-sensitive H^2 error is the largest violation; no successful long-distance neural result is established.

4.5 What the new results mean for the research goal

Derivative accuracy is the decisive new issue. Eleven of the 24 evaluated fits pass the earlier CPU pilot's L^2, H^1 and peak criteria, including all six evaluated test-sideband fits. Their individual `sampled_interval_pass` flags are true because that older flag excludes H^2. All 24 fail the long-distance gate that includes H^2; for those eleven, the extra derivative criterion causes rejection. Use the full campaign status, not the older flag, when reporting success. Initial gate violations occur between $\xi = 0.30$ and 1.10, although each evaluated fit was trained through 4.

This is directly relevant to the smoothness objective: a good envelope or peak prediction can conceal inaccurate derivatives. An H^2 error of 22.1% is an *approximation error relative to the reference*, not a 22.1% growth in the true norm and not evidence of a singularity. Its cause cannot be inferred from aggregate errors alone. Inspect per-mode error and residuals to distinguish spurious high-frequency content, truncation, phase drift and optimization error. The earlier CPU result and this run differ in inlets, horizon, network, bandwidth, budget and acceptance criteria; the change cannot be attributed to GPU execution.

Periodic qualification and isolated packets have different limits. The summary records reference qualification through 64, but all packet domain checks fail at 8. Thus the recorded periodic numerical evidence extends much farther than the declared doubled-domain agreement test, which passes only through 4. This is not a whole-line certificate. The detailed records are needed to distinguish finite-box interaction from failed refinement. Likewise, sensitivity qualification through 16 is not a bound on the size of amplification: it says the prescribed response comparisons passed. The actual norms, amplification curves and qualification margins are not in the uploaded summary.

Execution and provenance limits. All 24 completed PINN records include positive GPU allocations, with reported peak allocated tensor memory 94,955,520 bytes (90.56 MiB). Their median recorded optimization time is 62.61 seconds (range 47.63–67.44). This is not total device memory or a matched CPU/GPU speed comparison. CUDA 12.8 and PyTorch `2.10.0+cu128` are recorded. The exact GPU model, node and scheduler allocation are not in the published records; the declared `gpu2` campaign targets A100 hardware. Job and runtime-lock files are only hash-listed. The random-fit exception logs are absent, so neither memory exhaustion nor an optimizer failure can be asserted as their cause.

Next decision. Retrieve the detailed numerical `result.json` files, the PINN H^2 traces and `tasks/pinn-024.json` through `pinn-035.json` before diagnosing the missing random fits. Use validation cases to test shorter slabs, resolved bandwidth and optimization settings under controlled budgets. Do not automatically extend failed fits to larger distances. Freeze any selected repair before evaluation on a new untouched test set. These generic-wave results inform the learning method; they still do not test solitary-wave stability.

5 The solitary-wave implementation and a new resolution audit

The local solitary-wave code computes a conjugate-reflection-symmetric profile, then propagates it without enforcing that symmetry. A historical smoke at $\epsilon = 0.025$, $\mu = 2$, $s = 0$, $L = 160$, $N = 4096$ reached only $\xi = 0.1$. Its full profile residual was 2.03×10^{-11} and unperturbed final relative complex L^2 error was 7.09×10^{-11}. Its field checksum was verified against the stored report. This checks consistency over a short interval.

The standalone solitary files and their refinement configuration are local, untracked work, absent from the audited pushed tree. Their loader still depends on a historical profile file/sidecar even in recompute mode. The long-distance cluster campaign uses generic inlets, not solved solitary profiles. No completed refined long-distance solitary study, real-linear stability spectrum, or solitary-specific PINN/FNO result was found.

5.1 Fourier representation can defeat the test before training

On a periodic grid of length L, let $k_m = 2\pi m/L$ and $\widehat{Q}_m = \mathrm{FFT}(Q)_m/N$. Define

$$p_r(K)^2 = \frac{\sum_{|m|>K}(1+k_m^2)^r|\widehat{Q}_m|^2}{\sum_m(1+k_m^2)^r|\widehat{Q}_m|^2}. \tag{5.1}$$

Since Fourier modes are orthogonal in these discrete H^r norms, every approximation supported in $|m| \le K$ has relative error at least $p_r(K)$. This statement follows by splitting its squared error into retained and discarded Fourier modes. Phase and translation alignment preserve the support and cannot eliminate this floor.

K	$k_{\max}$	p_0 (%)	p_1 (%)	p_2 (%)
48	1.885	19.146	38.520	66.086
96	3.770	2.989	10.074	27.467
192	7.540	0.08096	0.50049	2.44868
256	10.053	0.007532	0.060796	0.386883
384	15.080	0.0000664	0.0007879	0.0073469

Table 3: New calculation on the stored $L = 160$ Dysthe profile; not a new propagation or neural-training experiment. The same-grid NLS seed also has a 59.77% H^2 floor at $K = 48$. Data and source hashes accompany the report.

The published neural cutoff $K = 48$ belongs to $L = 40$, where $k_{\max} = 7.540$. Copying the integer cutoff to $L = 160$ reduces physical bandwidth by a factor of four. The failure above is therefore a proposed-transfer problem, not an error in the original generic-packet results.

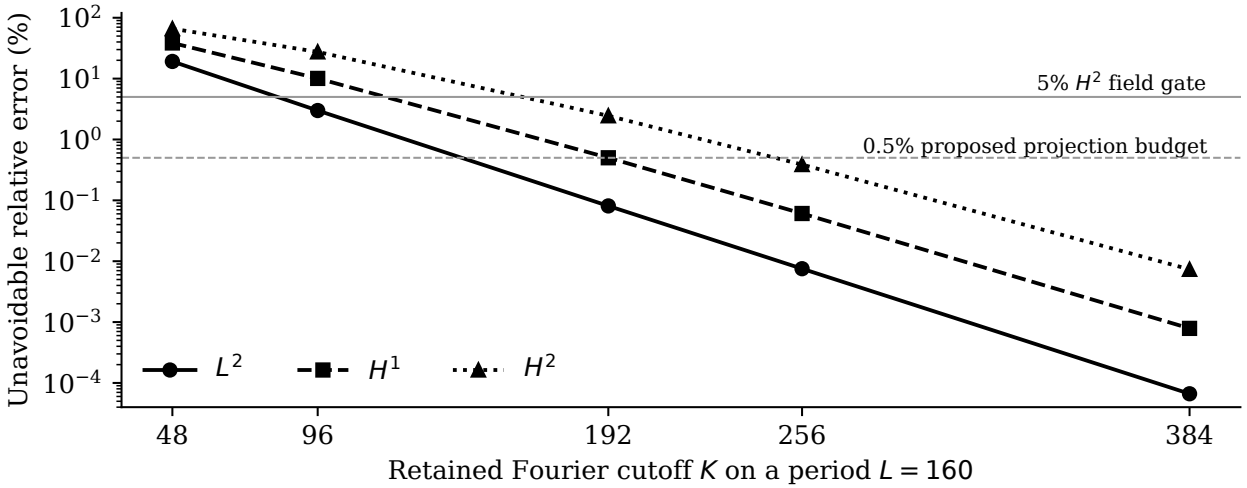


Figure 3: A lower bound on attainable errors for a strictly projected output. The 0.5% line is a proposed allocation of one tenth of the 5% H^2 field budget to inlet projection, not an existence criterion.

For this profile, $K = 256$ is a reasonable first *field-accuracy* trial under that projection budget, with at least 513 complex Fourier coefficients and an adequate sampling grid. A choice such as $N = 1024$ represents the profile band; full cubic residual quadrature needs more points, for example $Q_{\text{quad}} = 2048$. For a trigonometric output of band K, the cubic RHS has band at most $3K$; $Q_{\text{quad}} > 6K$ resolves its squared norm without Fourier aliasing. Audit with a larger quadrature and finer distance sampling. This is a starting design, not a measured memory requirement or sufficient accuracy guarantee for all profiles.

5.2 Two additional measurement gaps

The present solitary alignment minimizes L^2 distance, not H^1 distance. An H^1 orbit diagnostic must be added and verified before reporting it. Also, its perturbation coefficient multiplies a localized amplitude/phase factor; it is not itself the H^1 norm of the perturbation. Record the actual inlet norm or normalize directions. The generic long-distance sensitivity implementation already uses normalized directions, but that is a separate test.

Proposition 5.1 (Error in a signed neural response)**.** *For $\delta > 0$, let $w_\delta = (u_+ - u_-)/(2\delta)$ and $\widetilde{w}_\delta = (\widetilde{u}_+ - \widetilde{u}_-)/(2\delta)$, with all four fields in H^1. Then*

$$\|\widetilde{w}_\delta - w_\delta\|_{H^1} \le \frac{\|\widetilde{u}_+ - u_+\|_{H^1} + \|\widetilde{u}_- - u_-\|_{H^1}}{2\delta}.$$

Proof. Subtract the two responses and apply the triangle inequality in H^1. □

Thus a 1% field gate does not ensure accuracy for a perturbation of size 10^{-3}. Separately optimized PINNs may differ primarily because of fitting error. Test the response against the refined numerical response directly and report its error floor. A proposed 10% response gate needs an absolute fallback when the response is tiny; it is stricter information than passing the field gates.

6 A staged solitary-wave experiment with explicit decisions

The following extension is *proposed*, not already present in the pushed campaign. It begins with the existing $\mu = 2$, $s = 0$ branch and $\epsilon = 0.025, 0.05$. Broaden the branch only after the first cases qualify. Avoid claiming novelty for solitary-wave construction or collision inelasticity alone; both have established precedents.

S1. Construct and qualify the profile. *Goal:* distinguish an actual Dysthe traveling solution from an NLS guess. Continue from $\epsilon = 0$ and evaluate the full complex traveling residual on an oversampled grid. Refine profile grid and domain separately; recompute instead of padding. Record residual, projection floors, spectral and boundary tails, peak and mean flow. The local residual gate is 10^{-6}; require refinement agreement below the perturbation response. Failure means unresolved, not absent. The analytic NLS control is exact on the line; its periodic-box use needs domain checks. Refine propagation step $h/2$ separately in S2.

S2. Measure persistence and perturbation response. *Goal:* distinguish shape deformation from position and phase drift. Start with an unperturbed control and signed localized amplitude, phase and odd-amplitude perturbations. Normalize or report their actual H^1 sizes. Use sizes 10^{-3} and 5×10^{-4}, and checkpoints such as $\Xi = 1, 2, 4, 8, 20$, subject to reference/domain qualification. Save densely enough to resolve focusing and transient growth; add samples when a peak is missed. The evolved control measures numerical drift. The local plan saves only 21 stations through 20; increase sampling before interpreting event timing.

S3. Connect growth to a linearized mechanism. *Goal:* test whether observed perturbation growth agrees with a converged linear instability. In the moving/rotating frame the operator is real-linear:

$$\mathcal{A}v = DF(Q)v + i\mu v + s\partial_\tau v.$$

It includes variations through complex conjugation and the nonlocal mean flow. Verify the approximate neutral directions iQ and Q', then refine grid and domain and compare any positive growth rate with nonlinear propagation while the perturbation remains small. Purely imaginary computed eigenvalues do not prove nonlinear stability; transient growth and unresolved spectrum remain possible. This operator is not currently implemented.

S4. Test a neural method on already qualified cases. *Goal:* determine whether the selected slab PINN reproduces the numerical wave's drift, deformation and radiation. Set the Fourier cutoff from the profile and propagated spectral tails, rather than copying $K = 48$. Retain the original field gates, add phase/position observables, and apply a separate response gate before making sensitivity claims. Compare a shorter slab and higher bandwidth at matched, recorded optimization cost. Preserve every failed case and seed.

S5. Collisions and emergence, after S1–S4. *Goal:* reproduce known interactions before testing learned predictions. Compare separated Dysthe profiles with NLS at matched relative phase, separation and speed; measure outgoing profiles and radiation with box refinement. Motion in retarded time is not counterpropagation of physical carriers. Later, fit generic-packet peaks to the qualified branch and check traveling residual and lifetime. This does not establish a universal soliton-resolution conjecture.

7 What to measure and what counts as success

For v sampled on the periodic grid, use the consistent norms

$$\|v\|_{H^r}^2 = L\sum_m (1+k_m^2)^r |\widehat{v}_m|^2, \qquad r = 0, 1, 2.$$

For a fixed branch profile define the proposed orbit distance

$$d_1(u, Q) = \inf_{\theta, a} \frac{\|u - e^{i\theta} Q(\cdot - a)\|_{H^1}}{\|Q\|_{H^1}}. \tag{7.1}$$

The branch parameters and amplitude of Q stay fixed during this fit. The symmetry-orbit viewpoint is standard in solitary-wave stability theory [9], but that theory's assumptions are not automatically verified here.

Observable	Interpretation and safeguard
Raw complex error	Tests actual prediction, retaining phase and arrival error. Use alongside aligned error, never replace it by amplitude-only error.
d_1, fitted phase/shift	Measures deformation modulo symmetry; report phase and position separately so a wrong speed cannot be hidden by alignment.
Peak, width, spectral mean	Test coherent-wave amplitude, spreading and carrier shift. Specify how width and frequency are estimated and unwrap position.
Radiation measure	Report both fixed-window tail action and fitted-profile remainder. They are diagnostic definitions, not a unique nonlinear decomposition.
H^1, H^2 and spectral tails	Check derivative growth and whether higher modes remain resolved. Finite Fourier outputs are smooth by construction.
Correct invariants	Monitor action and the model's modified invariants. Conservation alone cannot establish accuracy or a coercive regularity bound.
Sensitivity amplification	Normalize initial perturbations, retain signs and check $\delta/2$. A few directions do not bound the full evolution operator.

Define an *observed persistence distance* as the last saved station such that all preceding saved orbit errors are below a predeclared tolerance (for example $d_1 \le 0.05$) and all numerical/domain checks pass. Report a wave that passes at the final station as "persists through Ξ"; the lifetime is right-censored, not infinite. A 5% threshold is an experimental choice, not a stability theorem or a universal water-wave validity boundary.

Numerical evidence for a loss of regularity would require growth that survives independent grid/step/domain refinement and increasing spectral bandwidth. A rising tail at the cutoff usually first establishes under-resolution. Distinguish that from optimization failure, boundary return and physical-model breakdown. Failure to observe growth likewise does not prove global smoothness.

8 How PINNs and FNOs should be used here

8.1 Slab PINN: a separately fitted solution on successive intervals

A slab PINN solves $[\xi_j, \xi_{j+1}]$, passes its predicted endpoint to the next slab, and repeats. The current implementation uses a network in ξ that outputs complex Fourier coefficients in τ; this enforces periodicity and permits complete-slice evaluation of $|D_\tau|(|u|^2)$. It does not sample the nonlocal term from disconnected points. Numerical labels are reserved for auditing; slab endpoints must not be reset to the reference during evaluation.

Sequence training has methodological support in PINN failure-mode work [10]. Wang et al.'s causal weighting is a related but distinct algorithm [11]: sequential slabs alone are not their specific residual weighting rule. Any added weighting should be a separately tested ablation. Residual minimization measures equation fit over the fitted interval, not forecast generalization or an existence certificate.

For solitary waves, first compare the existing spectral correction against a known moving/rotating reference representation, both with the same inlet and full residual. A moving frame may reduce phase/translation learning burden, but it must not freeze amplitude or suppress admissible perturbations. The benefit is a hypothesis to measure, not an established Dysthe result.

8.2 FNO: learn across initial functions, not one profile

FNO learns an operator from a family of functions [12]; duplicating one profile with many snapshots does not provide many independent inlets. A useful new dataset would span qualified branch parameters, global phase, translations, and controlled amplitude/phase/radiative perturbations, then include generic packet controls. Keep all snapshots, augmentations, signed perturbations and refinements of a parent inlet in one train/validation/test group.

Compare the same architecture trained with data loss against a PINO residual augmentation [13], using equal data, seeds and recorded cost. Freeze the choice on validation fields before test inspection. Evaluate short-step accuracy and autonomous rollout without reference resets. Separate interpolation within the sampled branch from held-out amplitude/steepness ranges. More epochs alone are not a supported remedy for the existing pilot failures.

An ordinary FNO's latent Fourier cutoff does not by itself strictly limit its output, because local nonlinear layers can create higher frequencies. The projection floor in Section 5 applies to the explicit output projector used in this project's spectral models. Record latent modes, output cutoff and grid separately. Phase/translation augmentation tests physical symmetries but should not inflate the independent test count.

8.3 Priority and cost

Run reference qualification and the slab PINN extension first. Treat FNO recovery as a separate controlled operator study. On A100, record data generation, optimization, validation, checkpoint I/O, evaluation and peak memory; compare total cost at matched measured accuracy against the Fourier solver. A cached neural forward pass is not the cost of obtaining a new per-inlet PINN solution.

9 Recommended research claim and verification record

The near-term question is:

> For a specified family of resolved spatial free-surface Dysthe solitary waves, how do perturbation growth, phase/position drift and radiation depend on steepness and propagation distance, and which of these behaviors can a slab PINN or an operator model reproduce within a measured error budget?

This can produce useful numerical evidence without promising a global existence proof. A subsequent proof project could study small-ϵ continuation from NLS or validated a-posteriori existence, but would need explicit function spaces, kernel removal, nonlocal estimates and validated inverse/tail bounds. Neither a converged iteration nor an ML residual supplies these ingredients.

Immediate changes justified by this review. Make the solitary workflow self-contained; add H^1 orbit fitting and normalized perturbation reporting; select bandwidth from projection/refinement evidence; increase event sampling; and keep response accuracy separate from field accuracy. The real-linear spectral operator is a later explicit implementation task. The report does not claim these proposed changes have been implemented or pushed.

Evidence inspected. The audit checked remote branch identities, the CPU-only CI record, published configs/results, local solitary source and the stored smoke checksum. Scientific CPU science trajectories were not rerun. New calculations covered Fourier projection and aggregate consistency. The exact published source passed all 39 software tests, including bounded training/restart checks, plus scope and artifact validation; see `data/remote-audit.json`. The new results commit changes no source. Its metadata were downloaded and hash-checked. No HPC job was submitted by this review and no cluster filesystem inspected. Detailed qualification, raw trajectories and failure logs are reported at the manifest's source location. NLS methodology is not relabeled as Dysthe validation.

Reproducible report package. `review.tex`, `royden-report.sty`, the three vector PDF figures, `make_figures.py`, and archived data are self-contained. Spectral powers reproduce (5.1); the JSON records the source NPZ checksum. `verification.json` records pdfLaTeX compilation and rendered-page checks. Preprint assumptions were inspected; proofs were not independently formalized.

Reviewed source identities.

- Experiment source: `c04b44ba6459cac6f216e3a59de4fac78e3b55d7`.
- Results publication: `723883f27a6b3489bfc36f55d9d15b52e8635ad0`.
- Core dependency: `a47a42321475feb6bc55031f4b752419f4b7fda9`.
- Learning dependency: `db7274f8cfe959cf187351f56b3dd4bf89f0e9b4`.

Historical solitary evidence remains distinct from those published commits. Its source NPZ has SHA256

`63e8636ed53c3183c6f5c2abb246ed4e62d0a97499c0f7029f104de198e21123`.

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